BA305 Workshop 1 Status Report — Fall 2026
Team name: A1
Members (full names): Justo Berisso, Jonathan Sze, Lucas Chen, Noah D'Costa
Dataset (number and name): Dataset #15 - Hotel booking records
Repository link: https://github.com/jberisso/BA305-A1---hotel-bookings

1. WHERE YOU ARE

We loaded the dataset and explored it to understand its structure and key variables. We analyzed each country by the number of bookings and grouped them into 8 primary countries and "Other" countries. We created visuals and analyzed the relationship between lead time and cancellations across different countries. We completed PCA to address our descriptive research question. After that, we completed k-NN for our predictive research question. Finally, at the end of our notebook, we implemented a leakage check by comparing assigned and reserved room types based on cancellation status.

2. WHAT THE DATA TURNED OUT TO LOOK LIKE

We found that our dataset contains bookings from 177 unique countries, with Portugal accounting for approximately 41% of all bookings. We found that 120 of the 177 countries had fewer than 50 bookings each, so we grouped these less-represented countries into an "Other" category. We observed that cancellation rates increase with longer lead times in Portugal, reaching approximately 88% for bookings made six or more months in advance. In contrast, most other countries have cancellation rates between 0% and 30%, while Spain and Italy generally fall between 40% and 55%. We also verified that these patterns were not simply the result of small sample sizes, as even the smallest country and lead-time group combinations contained more than 150 bookings. When we ran our PCA, which included lead time, average daily rate (ADR), number of adults, number of children, and special requests, it showed substantial overlap between city hotel and resort hotel bookings. This suggests that these features do not clearly separate the two hotel types into distinct clusters, apart from a small number of outlier bookings with extreme values. The first two principal components explained about 53% of the total variation, with the first accounting for about 30% and the second for about 22%. We also ran k-NN with an expanded feature set (lead_time, adults, country_grouped, market_segment, customer_type), and it reached 79% accuracy at k = 12. We performed a leakage check to identify suspicious relationships and found that the variable reservation_status revealed the outcome we were trying to predict, essentially giving the model the answer. We also plan to investigate whether assigned_room_type is recorded or updated before or after our intended prediction time. These changes will help refine our model, ensuring that it isn't being trained on data that it doesn't have access to when making our predictions.

3. CHANGES SINCE THE PROPOSAL

None

4. PROPOSAL FEEDBACK

Action 1: Expand predictive question beyond one variable

Status: Done. Our question was broadened from a single variable (deposit_type) to three (lead_time, country, market_segment) before this workshop, and we ran k-NN with an expanded feature set using lead_time, adults, country_grouped, market_segment, and customer_type.

Action 2: Experiment with including vs. excluding the suspicious booking agents

Status: not started / in progress. We identified the 81/87 agents with the 100%-cancellation pattern and flagged it as suspicious, but we haven't yet tried running analysis both with and without those rows to see if it changes anything.

5. BLOCKERS

We're not yet sure whether to treat Portugal's outsized share of the dataset (~41% of all bookings) as something to control for so the cross-country comparison is fair, or as a result worth investigating and reporting on its own. It's a decision that affects how we frame our descriptive findings. You can find this data when we analyze each country's bookings.

6. WHAT YOU WANT FROM US

Given that Portugal makes up 41% of the bookings and shows a much steeper cancellation-rate vs lead-time pattern than every other country, should that be treated as a confound to control for when comparing countries, or is the concentration and behavior worth highlighting as a finding?

We excluded the suspicious Non-Refund bookings (the 81/87 agents at 100% cancellation issue) in our lead-time trend analysis, but the k-NN model still includes all of them. Should that same cleaning decision be applied consistently across your whole analysis, including the predictive model, or is it fine to make that call analysis by analysis depending on what we're trying to show?